Bagaimana ACOSE dapat dilakukan? Pipeline Machine Learning vs Deeplearning vs LLM?

Implementasi ACOSE membandingkan **tiga paradigma komputasi utama**.

Implementasi *Aspect-Category-Opinion-Sentiment-Emotion* (ACOSE) berfokus pada tiga aspek utama: alur proses pipeline *machine learning* konvensional, model gabungan *deep learning*, serta kerangka generatif *Large Language Models* (LLM) (Zhang et al., 2022; Liu et al., 2020; Takallou et al., 2026).

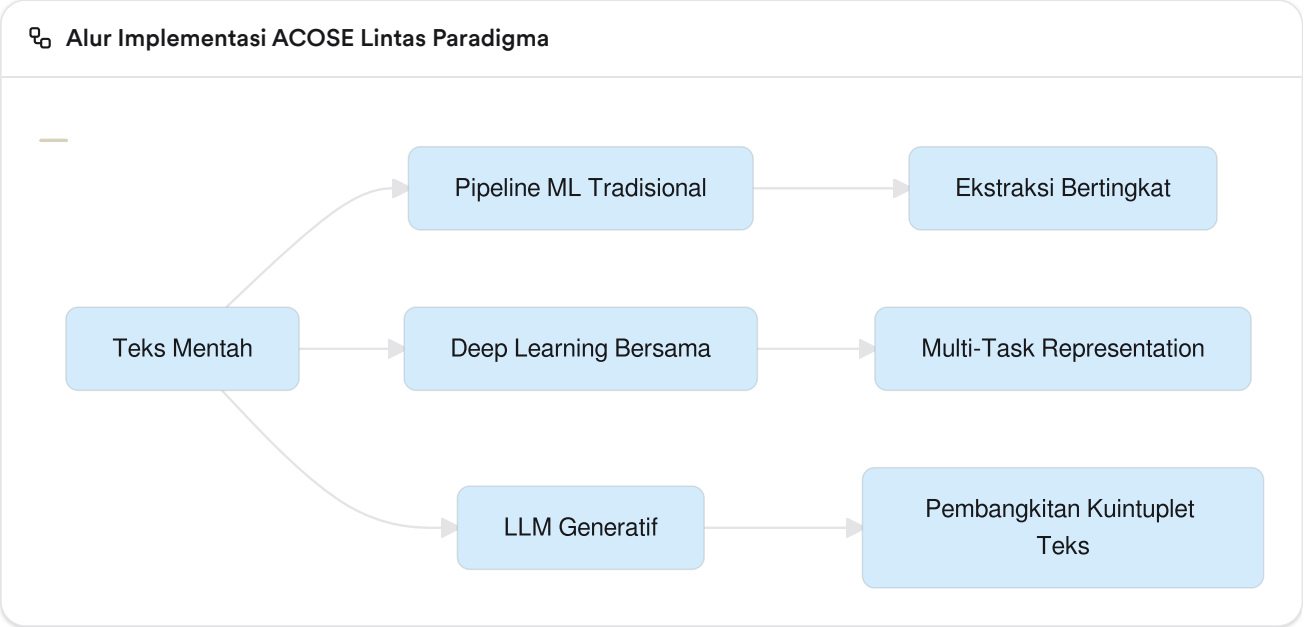


FIGURE 1 Komparasi alur kerja ekstraksi ACOSE menggunakan arsitektur ML, DL, dan LLM.

Pipeline Machine Learning Tradisional

Pendekatan *machine learning* (ML) tradisional memecah ACOSE ke dalam subtugas terpisah yang dieksekusi berurutan menggunakan rekayasa fitur manual (Zhang et al., 2022; H. et al., 2019; Hemmatian & Sohrabi, 2017).

- Metode pipeline menghubungkan modul ekstraksi entitas dan klasifikasi secara modular dengan memanfaatkan algoritma seperti pohon keputusan atau SVM linier (Zhang et al., 2022; Brauwers & Frasincar, 2021).
- Kinerja sistem sangat bergantung pada operasi pra-pemrosesan data seperti penyandian dan rekayasa fitur manual (Gregori et al., 2025).
- Kelemahan mendasar model bertingkat ini adalah kerentanan terhadap akumulasi perambatan galat (*error propagation*) dari langkah awal ke langkah hilir (Zhang et al., 2022; Xu et al., 2026).

Representasi Terpadu Deep Learning

Arsitektur *deep learning* (DL) mengatasi pemisahan fitur dengan menangkap representasi semantik dan sintaksis tingkat tinggi secara otomatis (Chauhan et al., 2023; H. et al., 2019; Brauwers & Frasincar, 2021).

- Model sekuensial seperti CNN, LSTM, dan Transformer memungkinkan ekstraksi elemen sentimen secara terpadu (*end-to-end*) (Truşcă & Frasincar, 2022; Ray & Chakrabarti, 2020; Brauwers & Frasincar, 2021).
- Mekanisme atensi bersama (*co-attention*) menghubungkan deteksi aspek implisit dengan polaritas sentimen dalam representasi laten gabungan (Truşcă & Frasincar, 2022; Brauwers & Frasincar, 2021).
- Kebutuhan data latih berskala besar serta sumber daya komputasi tinggi menjadi kendala utama pelatihannya dari awal (Brauwers & Frasincar, 2021).

Generasi Kuintuplet Menggunakan LLM

Model bahasa besar (LLM) mereformulasi ekstraksi elemen menjadi tugas pembangkitan teks bebas (*sequence-to-sequence*) yang terpadu (Zhang et al., 2024; Mughal et al., 2024; Jurado, 2024).

- Penggunaan *prompting* dan dekomposisi kalimat memungkinkan instruksi LLM memetakan kuintuplet afektif secara simultan (Said et al., 2025; Zhang et al., 2024; Mughal et al., 2024).
- Teknik *parameter-efficient fine-tuning* seperti QLoRA atau Adalora memfasilitasi adaptasi domain efisien di bawah keterbatasan perangkat keras (Zhang et al., 2024; Xu et al., 2026; Fang et al., 2024).
- Arsitektur hibrida yang menggabungkan representasi *encoder* BERT dengan inferensi generatif LLM terbukti menghasilkan performa paling optimal (Said et al., 2025; Zhang et al., 2024).

Secara komparatif, evolusi metodologi bergerak dari modularitas pipeline ML menuju penalaran konteks holistik berbasis LLM generatif terpandu (Zhang et al., 2022; Gu et al., 2024).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

Brauwers, G., & Frasincar, F. (2021). A Survey on Aspect-Based Sentiment Classification. *ACM Computing Surveys, 55*, 1 - 37. https://doi.org/10.1145/3503044

Chauhan, G., Nahta, R., Meena, Y., & Gopalani, D. (2023). Aspect based sentiment analysis using deep learning approaches: A survey. *Comput. Sci. Rev., 49*, 100576. https://doi.org/10.1016/j.cosrev.2023.100576

Fang, Z.-H., Lin, Z., Chen, Z., Chen, X.-H., Gao, Y., & Fang, Y.-G. (2024). Automated Federated Pipeline for Parameter-Efficient Fine-Tuning of Large Language Models. *IEEE Transactions on Mobile Computing, 25*, 8782-8797. https://doi.org/10.1109/tmc.2025.3649881

Gregori, L., Lazzaro, P., Lazzaro, M., Missier, P., & Torlone, R. (2025). An LLM-guided platform for multi-granular collection and management of data provenance. *Journal of Big Data, 12*. https://doi.org/10.1186/s40537-025-01209-3

Gu, Y., You, H.-Y., Cao, J., Yu, M.-R., Fan, H.-R., & Qian, S.-Y. (2024). Large Language Models for Constructing and Optimizing Machine Learning Workflows: A Survey. *ArXiv, abs/2411.10478*. https://doi.org/10.48550/arxiv.2411.10478

H., H., Prasad, P., Maag, A., & Alsadoon, A. (2019). Deep Learning for Aspect-Based Sentiment Analysis: A Comparative Review. *Expert Syst. Appl., 118*, 272-299. https://doi.org/10.1016/j.eswa.2018.10.003

Hemmatian, F., & Sohrabi, M. (2017). A survey on classification techniques for opinion mining and sentiment analysis. *Artificial Intelligence Review, 52*, 1495 - 1545. https://doi.org/10.1007/s10462-017-9599-6

Jurado, I. C. (2024). LLM Prompting Strategies in Automated Machine Learning. *2024 IEEE International Conference on Machine Learning and Applied Network Technologies (ICMLANT)*, 74-79. https://doi.org/10.1109/icmlant63295.2024.00018

Liu, H., Chatterjee, I., Zhou, M., Lu, X., & Abusorrah, A. (2020). Aspect-Based Sentiment Analysis: A Survey of Deep Learning Methods. *IEEE Transactions on Computational Social Systems, 7*, 1358-1375. https://doi.org/10.1109/tcss.2020.3033302

Mughal, N., Mujtaba, G., Shaikh, S., Kumar, A., & Daudpota, S. (2024). Comparative Analysis of Deep Natural Networks and Large Language Models for Aspect-Based Sentiment Analysis. *IEEE Access, 12*, 60943-60959. https://doi.org/10.1109/access.2024.3386969

Ray, P., & Chakrabarti, A. (2020). A Mixed approach of Deep Learning method and Rule-Based method to improve Aspect Level Sentiment Analysis. *Applied Computing and Informatics*. https://doi.org/10.1016/j.aci.2019.02.002

Said, M., Smairi, N., & Abadlia, H. (2025). LLM Powered DeepSeek and BiLSTM Pipeline Used in Aspect Based Sentiment Analysis. *2025 IEEE 37th International Conference on Tools with Artificial Intelligence (ICTAI)*, 813-817. https://doi.org/10.1109/ictai66417.2025.00118

Takallou, M. A., Gautam, A., & Shirolkar, P. (2026). LLMs for End-to-End Machine Learning: A Comprehensive Review. *IEEE Access, 14*, 47717-47738. https://doi.org/10.1109/access.2026.3678429

Truşcă, M. M., & Frasincar, F. (2022). Survey on aspect detection for aspect-based sentiment analysis. *Artificial Intelligence Review, 56*, 3797-3846. https://doi.org/10.1007/s10462-022-10252-y

Xu, C., Wang, M., Ren, Y., & Zhu, S. (2026). Enhancing Aspect-Based Sentiment Analysis in Tourism Using Large Language Models and Positional Information. *Tsinghua Science and Technology*. https://doi.org/10.26599/tst.2024.9010138

Zhang, W., Li, X., Deng, Y., Bing, L., & Lam, W. (2022). A Survey on Aspect-Based Sentiment Analysis: Tasks, Methods, and Challenges. *IEEE Transactions on Knowledge and Data Engineering, 35*, 11019-11038. https://doi.org/10.1109/tkde.2022.3230975

Zhang, Y., Xu, H., Zhang, D., & Xu, R.-F. (2024). A Hybrid Approach to Dimensional Aspect-Based Sentiment Analysis Using BERT and Large Language Models. *Electronics*. https://doi.org/10.3390/electronics13183724